THE COTANGENT COMPLEX AS THE SHEAF OF DIFFERENTIALS PUSHED FORWARD ALONG THE STRUCTURE MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays a chain of three equalities; the proof justifies the middle reduction by flatness and then matches the outer terms through the standard resolution.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The computation of the lower shriek along a standard resolution. Lemma lemma-pi-shriek-standard, used at proof lines 564-567. Identifies the lower shriek of the displayed sheaf with the definition of the cotangent complex.
- The definition of the cotangent complex of a ring map. Definition definition-cotangent-complex-ring-map, used at proof lines 567-568. Fixes the object being identified.
- Flatness of the sheaf of differentials on the category. used at proof lines 557-561. Makes the derived pullback equal to the ordinary pullback.
- The description of the ordinary pullback sheaf by the rule on objects. used at proof lines 561-563. Gives the value used in the final identification.

Source reference. Original definition 08PN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11031–11031. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 08PR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11034–11034. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08PT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11036–11036. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A \to B$ be a ring map. Let π and i be as in (08PR). There is a canonical isomorphism*

$$L_{B/A} = L\pi_!(Li^*\Omega_{\mathcal{O}/A}) = L\pi_!(i^*\Omega_{\mathcal{O}/A}) = L\pi_!(\Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B})$$

in $D(B)$.

Proof. For an object $\alpha : P \to B$ of the category $\mathcal{C}$ the module $\Omega_{P/A}$ is a free P-module. Thus $\Omega_{\mathcal{O}/A}$ is a flat $\mathcal{O}$-module. Hence $Li^*\Omega_{\mathcal{O}/A} = i^*\Omega_{\mathcal{O}/A}$ is the sheaf of $\underline{B}$-modules which associates to $\alpha : P \to A$ the B-module $\Omega_{P/A} \otimes_{P,\alpha} B$. By Lemma 08PT we see that the right hand side is computed by the value of this sheaf on the standard resolution which is our definition of the left hand side (Definition 08PN). □